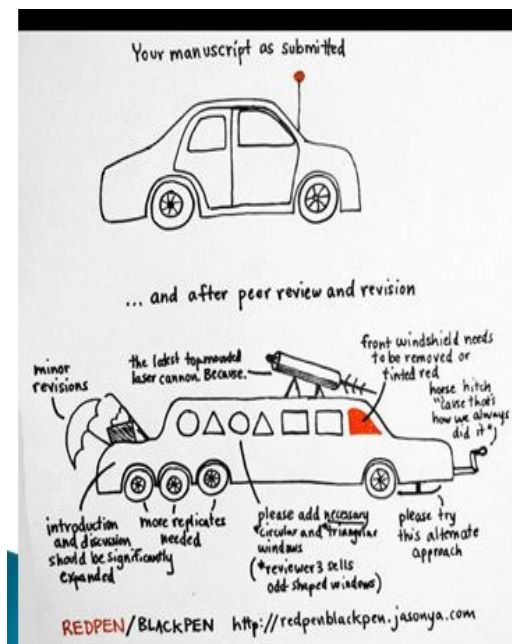


Swiss TPH Scientific writing

Lesson 2: Writing well-structured paragraphs and linking ideas

Lecturer: Riana Paola



1. Objectives:

- How to write paragraphs: paragraph structure, logical flow of ideas, using linking words
- How to establish cohesion & coherence
- How to connect sentences clearly.

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1. Paragraph structure

In English, the paragraph is the main unit of meaning. The rules for good paragraphs are strict. Readers expect writing to flow in logical paragraph units. Neglecting that structure may reduce the readability of your writing. This section focuses on constructing logical, coherent paragraphs.

1.1 Topic sentences/statements and supporting sentences

The topic sentence (also referred to as topic statement) expresses the main idea or topic of a paragraph, and often indicates how this idea will be developed. In academic and professional writing, the topic statement is most often the first sentence or first two sentences in the paragraph. Sometimes, however, the topic statement is the last sentence in a paragraph, e.g., when a paragraph builds up to a conclusion. A topic statement usually contains key words directly related to the topic. When it occurs at the beginning of a paragraph it also often suggests how the topic will be developed (by comparison and contrast, cause and effect analysis, etc.). Here is an example of an effective topic sentence and how it is developed in the paragraph.

Unlike gasohol-powered cars, fuel cell vehicles are virtually pollution free.* A methanol fuel system in a vehicle works through chemical reactions that leave the air clean. Firstly, a fuel processor breaks the methanol down into carbon dioxide and hydrogen. The hydrogen is then pumped into the cell itself, where it combines with oxygen to form water. This reaction causes electrons to be traded between molecules, which produces current that travels through an external circuit. The net products are thus carbon dioxide, water, and electricity. By contrast, when gasohol is burned in an internal combustion engine, it produces the same nitrous oxides that gasoline does.

(*Gasohol was defined as a blend of gasoline with at least 10% alcohol by volume, excluding alcohol made from petroleum, natural gas, or coal.)

The topic sentence in this paragraph is effective because it tells the reader what the main idea of the paragraph is. It also suggests the way in which the paragraph will develop, namely with a description of the fuel cell process and a comparison of its waste products with those of an internal combustion engine.

Skimming and scanning texts

In a text with well-structured paragraphs with clear topic sentences, readers can get the gist (central meaning) of the whole text by skimming it in this way:

1. Read the first paragraph and last paragraph in full.
2. Read the first sentences of the paragraphs in between.
3. Read headings, sub-headings, bullet points or key words (those in bold, italics, capitals or underlined) to guide you.

Skimming and scanning are essential reading skills for students who need to read many texts. **Scanning** is when you look over a text quickly, line by line, hunting for key words, dates, names and numbers. It's a useful skill to use when you need answers to specific questions.

A topic sentence is followed by several supporting statements. In texts with good readability, or *coherence*, these statements follow a consistent pattern of organization, one that flows naturally from the topic statement. Some of the most commonly used patterns of organization in academic and professional writing are chronological description, cause and effect analysis, comparison and contrast, listing, and general to particular ordering of details. Each of these patterns has certain characteristic features.

3.2 Cohesion

Cohesion refers to the way in which the ideas in a text are connected and the relationships between ideas are displayed. Cohesion is produced in formal writing mainly by using logical connectors and reference words.

Logical connectors establish clear connections between ideas and are thus essential to the successful communication of complex concepts and processes. The main types of logical connectors are linking words, which include conjunctive adverbs (including sequencers), coordinating and subordinating conjunctions and prepositions.

Reference words and phrases

Reference words and phrases refer backwards in the text and thereby allow ideas to be joined. Reference words and phrases either repeat or summarise previous information. Summary nouns / Summarizing nouns are good examples of ways in which to link ideas in texts.

3.2.1 Repetition

The techniques, reference words and phrases used for repetition are as follows:

1. simple repetition (i.e. the exact repetition of a word)
2. repetition with grammatical transformation (e.g. noun to verb)
3. demonstrative pronouns (especially 'this' and 'these')
4. personal pronouns (especially 'it' and 'they')
5. relative pronouns (which, that)
6. determiners (especially 'such', 'this', or 'these') followed by a previously used noun, called a summarizing / summary noun
7. synonyms

Synonyms are used to avoid simple repetition, but they cause the most problems for readers. If readers do not know that the synonym *is* a synonym, they will think that a new idea is being introduced. When you choose a synonym, make sure that it is the right one in that context.

Examples

1. The assembly machine uses robotic arms. The *arms* (simple repetition) can be telescopic or of a fixed length.
2. A good paragraph is designed to be easily readable. *Readability* (repetition with grammatical transformation) has several aspects.
3. The assembly machine uses robotic arms. *These* (demonstrative pronoun) can be telescopic or of a fixed length.
4. First of all, the samples were weighed. *They* (personal pronoun for 'samples') were then placed in an acid bath.
5. A solution was required *that* fulfilled all of the above requirements (relative pronoun).
6. After extraction, metals often undergo a number of further production processes. *These processes* (determiner plus repeated noun – see 'summarizing nouns' below) result in materials with significantly superior characteristics.
7. The project group had to deal with several problems. However, the *difficulties* (synonym for 'problems') were overcome and the project was completed on time.

3.2.1 Summarizing nouns aid cohesion

A summary noun summarises an idea which has been expressed earlier in a paragraph, either within the same sentence or in a previous sentence. Use this/these followed by a noun or phrase that summarizes the content of the previous sentence.

Examples:

- ▶ In recent years, the number of students applying to degree programmes has increased steadily, while the number of places has remained constant. *This situation* has resulted in intense competition for admission.
- ▶ Our pilot study has shown that wind turbines used to generate electricity can pose a threat to flying birds. *This finding* suggests a need for further research on improving the safety of these mechanisms.
- ▶ However, in the 1980s the debt crisis and the ensuing contraction of budgets prompted a re-appraisal of public sector provision. Donors began lobbying for the restructuring of public services; by the 1980s, they were demanding full-scale privatization. However, implementation of *such reforms* has been slow. (The use of the summary noun 'such reforms' summarizes the ideas 're-appraisal of public sector' and 'restructuring of public services'.)

3.2.2 Use linking words / conjunctions to improve cohesion.

There are more and better linking words than 'and', 'but' and 'because'!

Here are some of the most frequently used linking words/phrases:

- *Addition*: in addition, moreover, also, as well as, furthermore, too, also, not only... but also, and
- *Order/Sequence*: at first, to begin with, first, firstly, in the first place, second, secondly, in the second place, next, thereafter, following that, subsequently, later, eventually, lastly, finally, in the end (Note the difference in meaning between 'at first' and 'first':

- *Reason/justification:* Because/as/for/since, in order to, so as to, so that, accordingly, as a consequence/ consequently, as a result, for this reason, hence, owing to (this), so, therefore, thus
- *Contrast:* However, in contrast, still, by comparison, instead, even though/although, in contrast, on the other hand, nevertheless, nonetheless, conversely, yet, albeit (pronounced 'all-bee-it')
- *Conclusion and summary:* and so, finally, in brief, in closing, in conclusion, on the whole, to conclude, to summarize
- *Example words/Illustration:* as an example, for example, for instance, specifically, thus, to illustrate
- *Restatement words:* in other words, in short, in simpler terms, that is, *i.e.*, to put it differently, to repeat
- *Insistence words:* in fact, indeed
- *Purpose:* to, in order to, so as to (Note the **word order of the negative forms 'in order not to' and 'so as not to'**)
- *Time words:* when, whenever, while, before, after, until, as soon as, once, by the time, since, in the meantime

Note the **use of commas** with linking words! With certain linking words such as *however, therefore, besides, thus, finally, etc.* one needs to use punctuation marks. Look at this example:

- The findings of the report revealed shortcomings in the current system; *therefore*, other methods should be investigated or developed.

In the case of linking words and phrases that are used to express an additional thought, one should add a semi-colon after the first complete thought, and a comma after the linking word or phrase, such as in the example above. Another possibility is simply to separate the two sentences with a full stop, for example:

- The findings of the report revealed shortcomings in the current system. *Therefore*, other methods should be investigated or developed.

Note the **sentence construction with parallel structures** such as 'either...or', 'neither ... nor', and 'not only... but also...': both parts of this connector should be followed by the same part of speech, as in the examples below.

- Leonardo da Vinci was not only an inventor (noun) but also an artist (noun).

If you want to add extra emphasis, these connectors could be put at the beginning of a sentence.

- Not only did (verb) Chris Barnard succeed in successfully transplanting the first human heart, but he also researched (verb) the use of artificial heart valves.

(The same sentence construction is applied to sentences with 'either... or...' and 'neither... nor...')

Consult https://www.grammarly.com/blog/parallelism-with-not-only-but-also/?gclid=Cj0KCQiAs5eCBhCBARIsAEhk4r5MOQmFPvntaWXZeXVrPubmr3N4gLU3CqhL_9wvdd7sRkYxl-ZifmEaAtzHEALw_wcB&gclidsrc=aw.ds for a more comprehensive explanation of such parallel structures.

Example text showing cohesive devices

In the following text, the logical connectors are marked in *purple italics* and the reference/linking words and phrases are marked in **bold**.

Structures which are most at risk from progressive failure are **those** that are highly optimized and operate at a high proportion of their maximum load. Aerospace frames are perhaps the most obvious example of **such structures** and it is in the design of **these frames** that much attention has been paid to ensuring fail-safety. *Since they* must have as low a weight as possible, **aerospace frames** are **optimized** to ensure that no part is larger than necessary. *However, this* means that a variation from the assumed **load** pattern, which would occur in the case of local damage, could have a serious effect on members close to the site of **damage**. *Therefore*, allowance must be made for **this possibility**. Fail-safe or damage-tolerant design is **that** which preserves the serviceability of **members**, *even after they* have been **damaged**. The main method by which **this** is achieved is the incorporation of alternative **load** paths into **the structure**. **These alternative paths** produce multiple redundancy; *thus*, when **damage** occurs, there are several other **members** that can carry the extra **load**.

3.2.3 Sequencers

Sequencers are used to connect items in a list or steps in a process. Writers often use them in when describing the methods used to investigate something, for example when writing the methods sections of reports or research articles. The most common sequencers are as follows:

- First, / Firstly, / First of all,
- Second, / Secondly,
- Third, / Thirdly,
- Fourth, / Fourthly,
- Next,
- Then (usually without a comma) / Subject (+ auxiliary verb) + then + main verb (e.g. We then collected samples...)
- After that, / After + noun phrase or clause (usually in the present perfect or past perfect) (e.g. After that, the participants donated samples...)
- When / As soon as / Once + clause (usually in the present perfect or past perfect) (e.g. As soon as the maximum temperature was reached, ...)
- While – for parallel events or actions (e.g. While one worker attaches the doors of the car, another fits the lights.)
- As – for closely connected actions or events, e.g. 'As the temperature rises, the liquid evaporates'. (Note: in formal English, 'as' can be a synonym for 'because', e.g. 'As the technique failed, the team had to find a new solution'.)
- Finally,

'Next', 'then', 'after that', 'after', 'when', 'as soon as', 'once' and 'and' can be used for any of the intermediate steps in a procedure. These sequencers are often used to give a description a better flow, sometimes by joining steps into one sentence. Compare these two descriptions:

A) First, the specimens were collected. Second, they were labelled. Third, they were analysed. Fourth, the results were recorded. Next, a report form was completed. Then the report was sent to the chief engineer. After that, the report was read. Finally, the report was acted upon.

B) First, the specimens were collected and labelled. They were then analysed and the results recorded. After that, a report form was completed and sent to the chief engineer. Finally, the report was read and acted upon.

The conjunction 'and' is used to join closely related steps into a single sentence, e.g. 'The subjects were selected at random and (were) interviewed by telephone'. Using 'and' in this way not only improves flow, but also allows the description of a procedure to be broken down into sub-steps. This structural arrangement is especially useful in longer descriptions. However, do not use 'and' to string many ideas together, as the reader then might lose track of what is connected in the string.

Assignment 2 (Two parts, A and B)

A. Choose 3 of the topic sentences below and develop each of them into a cohesive, coherent paragraph of about 100-120 words, making sure that the supporting sentences are well linked to the topic sentence and successive sentences. Use the techniques and linking words in the script and slides to improve cohesion and coherence. Reference all sources used in the text and in a short bibliography. Put your paragraph through the plagiarism check, Turnitin, on the ADAM page. Upload the Turnitin report too (not the receipt). (See instructions above.)

1. Primary Healthcare (PHC) is made up of three main areas: empowered people and communities; multisectoral policy and action; and primary care and essential public health functions as the core of integrated health services.
2. Tuberculosis is a serious infection that affects the lungs and sometimes other parts of the body, such as the brain, bones, joints and kidneys.
3. Apart from well-known tropical diseases, such as malaria, lesser-known diseases also require control and treatment.
4. Climate change is expected to negatively impact health and wellbeing in all countries of the world.
5. Diarrheal diseases account for 1 in 9 child deaths worldwide, making diarrhoea the second leading cause of death among children under the age of 5.
6. The prevalence of obesity has increased worldwide in the past 50 years, reaching pandemic levels.
7. At the beginning of the 1980s, before HIV had been identified as the cause of AIDS, the infection was thought to only affect specific groups, such as gay men in developed countries and people who inject drugs.

B. Write a bullet-point checklist of at least 5 points to focus on when reviewing your peer's writing.

Name the assignment as follows: 2Surname name and upload it onto ADAM.

Then, choose your best short paragraph in your opinion and upload it to the Padlet page of this course by the same due date.